NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Cloud Computing

Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Mobile Cloud Computing (MCC) benefits mobile devices by:

1. Offloading compute-intensive tasks to the cloud
2. Extending effective battery life by reducing on-device processing
3. Guaranteeing zero network latency
4. Providing access to large-scale storage

A) 1, 2, and 4 only
B) 1 and 3 only
C) 2 and 3 only
D) All of the above

Answer: A
Solution: MCC offloads heavy computation (1), which helps conserve battery (2), and provides scalable cloud storage (4). It cannot guarantee zero latency (3), since wireless communication always introduces some delay.

QUESTION 2:

A city planning department stores road networks, building locations, GPS coordinates, and satellite images in a cloud-based system for analysis.

This is primarily an example of:

A. Mobile Cloud Computing

B. Spatial Cloud Computing

C. Serverless Computing

D. Batch Computing

Correct Answer: B

Detailed Solution:
Spatial Cloud Computing focuses on managing and processing geospatial information such as coordinates, maps, GIS layers, and satellite imagery.

QUESTION 3:

Which task is MOST likely to rely on Spatial Cloud Computing?

A) Sorting a company's payroll spreadsheet

B) Real-time flood-risk mapping using terrain and rainfall data across a region

C) Compiling source code

D) Streaming a music playlist

Answer: B
Solution: Region-scale flood-risk mapping combines terrain, rainfall, and location data — large-scale geospatial analytics well suited to spatial cloud systems. The other tasks involve no spatial data.

QUESTION 4:

In Cyber-Physical Systems (CPS), real-time feedback between sensors and actuators is essential for:

A) Data archiving
B) Closed-loop control
C) Static data analysis
D) Batch processing

Correct Answer: B

Detailed Solution: Cyber-Physical Systems tightly integrate sensing, computation, and actuation. Sensors collect data from the physical world, computation processes it, and actuators respond — forming a closed-loop control system, which is a core principle of CPS.

QUESTION 5:

A GPS-enabled taxi periodically records its location as: (x1,y1,t1),(x2,y2,t2),…,(xn,yn,tn)
What does this sequence represent?

A. Spatial trajectory

B. Spatial database

C. Road topology

D. Spatial index

Correct Answer: A

Detailed Solution:
A spatial trajectory represents the movement of an object through geographical space as a sequence of time-ordered spatial positions. Such trajectories are useful in transportation and mobility analysis.

QUESTION 6:

_____ refers to outsourcing location-based sensing or data-collection tasks to a crowd of mobile users who gather data at specific places and times.

A. Spatial trajectory

B. Spatial crowdsourcing

C. Time series

D. Road network graph

Answer: B

Solution: Spatial crowdsourcing assigns location- and time-specific sensing tasks to mobile participants (e.g., reporting road conditions or air quality at given spots). A spatial trajectory is instead a moving object's time-ordered path, not a task-assignment paradigm.

QUESTION 7:

In V2V communication, the primary objective is:
A) Providing internet access to vehicles
B) Exchanging safety-critical information such as speed, position, and braking

C) Supporting infotainment systems
D) Uploading vehicle data to cloud storage

Correct Answer: B

Detailed Solution: V2V focuses on direct vehicle-to-vehicle safety messaging (speed, heading, braking, lane changes) to prevent collisions. It is not meant for entertainment or internet access.

QUESTION 8:

A connected-vehicle system where onboard sensors capture data, roadside units preprocess and react locally, and the cloud performs fleet-wide long-term analytics is BEST described as:

A) Cloud-only computing

B) Edge computing with no cloud

C) An integrated Edge–Fog–Cloud architecture

D) A single centralized server

Answer: C

Solution: Onboard sensors (edge), roadside units doing local preprocessing/reaction (fog), and the cloud handling long-term fleet analytics distribute computation across three layers — an integrated Edge–Fog–Cloud architecture, not a cloud-only or edge-only system.

QUESTION 9:

A major benefit of real-time mobility analytics in urban transport systems is:
A) Eliminating the need for traffic laws
B) Dynamic traffic signal optimization based on live vehicle density
C) Reducing vehicle manufacturing costs
D) Preventing all road accidents

Correct Answer: B

Detailed Solution: Real-time analytics uses live data from sensors, cameras, and GPS to adapt traffic signals and routes dynamically, reducing congestion and delays.

QUESTION 10:

A multimedia application needs to deliver extremely high-resolution video and augmented-reality content to mobile users. Which 5G service category is most directly suited to this requirement?

A. eMBB

B. mMTC

C. URLLC

D. NFC

Correct Answer: A

Detailed Solution: Enhanced Mobile Broadband (eMBB) is designed to provide very high data rates and improved bandwidth efficiency. It is therefore well suited to applications such as high-resolution video and AR/VR streaming.